Ahmad Alfan Alfian Irfan

Software Developer & AI Research Intern

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Profile

Information Technology student with experience developing web applications, automation tools, and infrastructure systems. Research interests include decision-making under uncertainty, AI verification and validation, and explainable machine learning. Builds practical software for businesses, education, and research.

Experience

Research Intern — AI V&V Lab

June 2026 – present

Contributes to collaborative research on AI evaluation, verification and validation, and decision-making systems. Co-author of work on compressed driving policies, stochastic navigation, and machine-readable V&V specifications.

Research Intern — OISL

Healthcare AI research at Orthohealth Intelligence System Lab, including co-authorship of Knee-xRAI for explainable knee osteoarthritis grading. Developed the laboratory website.

Intern — PLN Batam

August – September 2025

Developed a substation management system for operational monitoring. Worked with internal teams to align data handling and system workflows with operational needs.

Technical Projects — KMTI UMY

November 2024 – present

Built a web-based payment tracking system and chatbot automation for student organization workflows. Coordinated technical tasks and maintained system documentation.

Selected Web Projects

Kopi Ngaji — Local e-commerce

Developed a storefront supporting a local coffee business and digital transaction management.

Berkah Sekawan Tangguh — Company profile & e-commerce

Built a company website and e-commerce platform for building materials.

Bimbel Aksara — Education platform / Capstone project

Combined a learning management system, information system, and company profile.

Merzzmail — Mail infrastructure & OISL — Research lab website

Developed a mail server project and a web presence for a research laboratory.

Technical Skills

Development: JavaScript, TypeScript, PHP, Python, Go, HTML/CSS, React, Next.js, Node.js, Laravel, Flutter, Kotlin, REST APIs.
Data & infrastructure: MySQL, PostgreSQL, Docker, Git, system design, automation, machine learning.

Research Publications

HiddenCourtesyMerge-Sim: A POMDP Benchmark for Bayesian Hidden-Courtesy Inference in Closed-Loop Merge Negotiation

Ahmad Alfan Alfian Irfan, Slamet Riyadi.
2026 International Electronics Symposium (IES), pp. 1–6. Conference paper.
Publication profile: Google Scholar

Preprints

A Closed-Loop Evaluation of Capability Loss and Recovery in Compressed Driving Policies

Ahmad Alfan Alfian Irfan, Nur Ahmad Khatim, Mansur Arief.
September 2026. arXiv:2609.00718 · Preprint.

Dijkstra as an Oracle for Online Stochastic Shortest Path Navigation with Provable Guarantees

Mansur M. Arief, Ali Akarma, **Ahmad Alfan Alfian Irfan**.
August 2026. arXiv:2608.17703 · Preprint.

Integrating High-Level Requirements to Low-Level Tests with Machine-Readable V&V Specifications

Mansur Arief, Nur Ahmad Khatim, Ali Akarma, **Ahmad Alfan Alfian Irfan**.
July 2026. arXiv:2607.17686 · Preprint.

Knee-xRAI: An Explainable AI Framework for Automatic Kellgren-Lawrence Grading of Knee Osteoarthritis

Azmul A. Irfan, Nur Ahmad Khatim, **Alfan Alfian Irfan**, Achmad Zaki, Erike A. Suwarsono, Mansur M. Arief.
April 2026; revised June 2026. arXiv:2604.23435 · Preprint.

Education

Universitas Muhammadiyah Yogyakarta

September 2023 – present

Bachelor’s studies in Information Technology.

Additional Projects

Binakata: Integrated dyslexia learning and early detection platform (2024–2025).
MedSync: Referral automation and dynamic patient queue management (2025).
OceanCare: Marine conservation platform combining literacy, action, and funding with image classification and serious game concepts (2025).

Awards & Certifications

- 1st place — UI/UX Competition FAST UTM, 2024.
- 3rd place — National Hackathon APTIKOM, 2025.
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate.
- Model Context Protocol: Advanced Topics — Anthropic, 2025.

Languages: Indonesian, English, Arabic.

Updated October 2026. Research preprints are listed separately from conference publications.